Rutgers University

Math 01:640:252 — Differential Equations

Harmonic Oscillators & The Hydrogen Atom

From Classical Mechanics to Quantum Wave Functions

Daksh Kaul

April 17, 2026

Chapter 1

Introduction

Oscillatory systems appear across nearly every physical domain, and the mathematics used to describe them draws from a common set of differential equation techniques regardless of whether the system is classical or quantum. This work traces that progression explicitly, beginning with the classical harmonic oscillator as a familiar two-component linear system with complex eigenvalues, then introducing the Schrödinger wave equation and the operator framework that connects physical observables to differential operators, working through the quantum harmonic oscillator as the first instance of more sophisticated solution techniques applied to a physically constrained eigenvalue problem, and finally applying the full separation of variables and asymptotic analysis to the hydrogen atom.

Furthermore, this work also emphasizes how the implications of these solutions extend beyond their immediate mathematical setting. The asymptotic behavior of each quantum system as the quantum number increases is analyzed and visualized to examine the physical consequences of this regime. In this limit, the quantum and classical descriptions of a physical system are expected to converge, and it is through this convergence that the classical picture emerges naturally as a limiting case of the quantum description, rather than as an independent framework.

Chapter 2

Classical Harmonic Oscillator

In classical mechanics the harmonic oscillator is defined as (LibreTexts Physics, 2025):

$$x(t) = A\cos(\omega t + \phi) \tag{2.1}$$
$$x'(t) = -A\omega\sin(\omega t + \phi) \tag{2.2}$$

where for an system governed by an elastic restoring-interaction $F_x = -kx$, along side its associated kinetic term $T_x = \frac{1}{2}m\dot{x}^2$ the total system energy is described by the Hamiltonian:

$$H = T + V \tag{2.3}$$
$$H = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}kx^2 \tag{2.4}$$

Written as a two-component system:

$$x' = \partial_p H \qquad p' = -\partial_x H \tag{2.5}$$

Since $p = mv \ldots$

$$H(x,p) = \frac{p^2}{2m} + \frac{1}{2}kx^2 \tag{2.6}$$

Therefore:

$$x' = \partial_p H = \frac{p}{m} \qquad p' = -\partial_x H = -kx \tag{2.7}$$

$$\begin{cases} x' = v \\ v' = \dfrac{p'}{m} = -\dfrac{k}{m}x \end{cases} \implies \begin{cases} x' = v \\ v' = -\dfrac{k}{m}x \end{cases} \tag{2.8}$$

Solving the System

$$Y' = AY \qquad \begin{bmatrix} x' \\ v' \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -k/m & 0 \end{bmatrix} Y \tag{2.9}$$

The characteristic equation:

$$\lambda^2 - \mathrm{Tr}(A)\lambda + \det(A) = 0 \tag{2.10}$$

$$\lambda^2 + \frac{k}{m} = 0 \tag{2.11}$$

$$\lambda^2 = -k/m \tag{2.12}$$

$$\lambda = \pm i\sqrt{k/m} \tag{2.13}$$

The corresponding exponential solution follows directly from Euler's formula:

$$e^{\pm i\sqrt{k/m}\,t} = \cos\Big(\sqrt{k/m}\,t\Big) \pm i\sin\Big(\sqrt{k/m}\,t\Big) \tag{2.14}$$

Eigenvectors

$$(A - \lambda I)|v\rangle = |0\rangle \tag{2.15}$$

$$\left(\begin{bmatrix} 0 & 1 \\ -k/m & 0 \end{bmatrix} - \begin{bmatrix} i\sqrt{k/m} & 0 \\ 0 & i\sqrt{k/m} \end{bmatrix}\right)|v\rangle = |0\rangle \tag{2.16}$$

$$\begin{bmatrix} -i\sqrt{k/m} & 1 \\ -k/m & -i\sqrt{k/m} \end{bmatrix}|v\rangle = |0\rangle \tag{2.17}$$

This gives the system:

$$-i\sqrt{k/m}\,x_1 + v_2 = 0 \tag{2.18}$$

$$-k/m\,x_1 - i\sqrt{k/m}\,v_2 = 0 \tag{2.19}$$

$$v_2 = i\sqrt{k/m}\,x_1 \qquad \text{assuming } x_1 = 1\ldots \tag{2.20}$$

$$v_2 = i\sqrt{k/m} \tag{2.21}$$

Therefore:

$$|v_1\rangle = \langle 1,\ i\sqrt{k/m}\rangle \tag{2.22}$$

For the second eigenvector ($\lambda = -i\sqrt{k/m}$), we can simply apply the complex conjugate therm by which any given characteristic polynomial of complex roots has $|v_2\rangle = |v_1^*\rangle$.

$$|v_2\rangle = \begin{bmatrix} 1 \\ -i\sqrt{k/m} \end{bmatrix} \tag{2.23}$$

General Solution

The two straight line solutions are:

$$Y_1(t) = e^{i\sqrt{k/m}\,t}\begin{bmatrix}1\\ i\sqrt{k/m}\end{bmatrix} \qquad Y_2(t) = e^{-i\sqrt{k/m}\,t}\begin{bmatrix}1\\ -i\sqrt{k/m}\end{bmatrix} \tag{2.24}$$

The general solution to the classical harmonic oscillator:

$$\begin{aligned} Y(t) &= k_1\, e^{i\sqrt{k/m}\,t}|Y_1\rangle + k_2\, e^{-i\sqrt{k/m}\,t}|Y_2\rangle & (2.25)\\ &= k_1\, e^{i\sqrt{k/m}\,t}\begin{bmatrix}1\\ i\sqrt{k/m}\end{bmatrix} + k_2\, e^{-i\sqrt{k/m}\,t}\begin{bmatrix}1\\ -i\sqrt{k/m}\end{bmatrix} & (2.26)\end{aligned}$$

Since the physical observable $x(t)$ represents a measurable displacement in real space, it must take real values for all t. The general eigenvector-based solution is constructed in $\mathbb{C}$ due to the complex conjugate eigenvalues $\lambda = \pm i\sqrt{k/m}$, so the coefficients must be chosen so that the imaginary components cancel in the superposition, ensuring $x(t) \in \mathbb{R}$ for all time. This requires $k_2 = k_1^*$. Writing $k_1 = \frac{A}{2}e^{i\phi}$ and thus $k_2 = \frac{A}{2}e^{-i\phi}$, the position component becomes:

$$\begin{aligned} x(t) &= \frac{A}{2}e^{i\phi}\, e^{i\sqrt{k/m}\,t} + \frac{A}{2}e^{-i\phi}\, e^{-i\sqrt{k/m}\,t}\\ &= \frac{A}{2}e^{i(\sqrt{k/m}\,t+\phi)} + \frac{A}{2}e^{-i(\sqrt{k/m}\,t+\phi)}\\ &= A\cos\Big(\sqrt{k/m}\,t+\phi\Big) & (2.27)\end{aligned}$$

where the last step follows directly from Euler's formula. The velocity is then:

$$\dot{x}(t) = -A\sqrt{k/m}\,\sin\Big(\sqrt{k/m}\,t+\phi\Big) \tag{2.28}$$

Substituting directly into the Hamiltonian to evaluate the total system energy:

$$\begin{aligned} H &= \frac{1}{2}m\dot{x}^2 + \frac{1}{2}kx^2\\ &= \frac{1}{2}m\cdot\frac{k}{m}A^2\sin^2\Big(\sqrt{k/m}\,t+\phi\Big) + \frac{1}{2}kA^2\cos^2\Big(\sqrt{k/m}\,t+\phi\Big)\\ &= \frac{1}{2}kA^2\Big[\sin^2\Big(\sqrt{k/m}\,t+\phi\Big) + \cos^2\Big(\sqrt{k/m}\,t+\phi\Big)\Big]\\ &= \frac{1}{2}kA^2 & (2.29)\end{aligned}$$

where $m\omega^2 = k$ was used to simplify the kinetic term. The total energy $E = \frac{1}{2}kA^2$ is therefore exactly constant at every instant, as required by energy conservation, and confines the evolution of the system between the turning points $x = \pm A$.

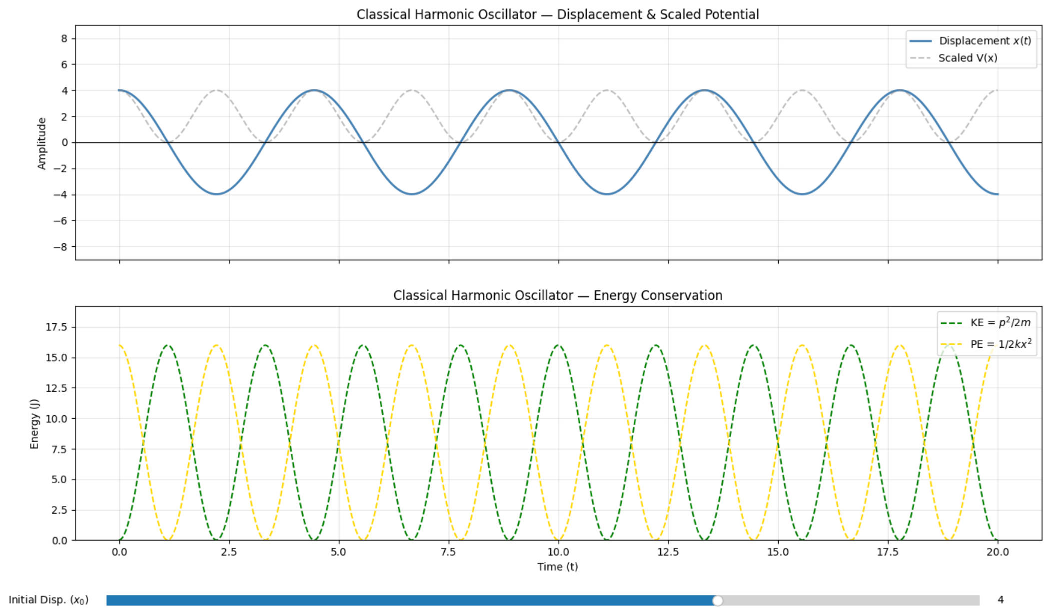

Figure 2.1: A classical harmonic oscillator set at an initial displacement of $x_o = 4$ displacement units.

2.1 Implications & Asymptotic Analysis

The classical harmonic oscillator moves sinusoidally between the turning points $x = \pm A$, with total mechanical energy conserved at all times as $E = \frac{1}{2}kA^2$. As it evolves, kinetic energy $T = \frac{1}{2}m\dot{x}^2$ and potential energy $V = \frac{1}{2}kx^2$ continuously convert into one another. At the turning points $x = \pm A$, the particle momentarily comes to rest, so all of the energy is stored as potential energy and the kinetic energy is zero. At the equilibrium position $x = 0$, the situation is reversed: the potential energy vanishes and the speed, and therefore the kinetic energy, is maximal. The motion is thus a smooth exchange between these two forms of energy, fully constrained by conservation of energy and determined entirely by initial conditions.

This behavior is governed by the angular frequency $\omega = \sqrt{k/m}$. While the total energy $E = \frac{1}{2}kA^2$ depends only on the spring constant k and the amplitude A, it is independent of the mass. The mass instead controls how quickly the oscillation occurs. A larger mass increases the system's inertia, so it responds more slowly to the restoring force, leading to a smaller ω and a longer period $T = 2\pi\sqrt{m/k}$. On a graph of $x(t)$, this appears as a sinusoid stretched along the time axis: the amplitude remains the same, but the peaks and troughs are spaced further apart. The same idea carries over to the energy exchange: the maximum values of kinetic and potential energy remain fixed by $E = \frac{1}{2}kA^2$, but the rate at which energy shifts between them depends on the mass.

Chapter 3

The Schrödinger Wave Equation

In Quantum Mechanics, it is postulated that given any isolated quantum system its complete physical state can be described by a complex-valued 'wave-function' commonly defined $\Psi(r,t)$ (Weichmann, 2024), that is defined within a Hilbert Space (LibreTexts Engineering, 2022).

Where the time-independent Schrödinger equation is as follows:

$$-\frac{\hbar^2}{2m} f\,\nabla^2\psi + V(x)\psi f = i\hbar\psi f' \tag{3.1}$$

$$-\frac{\hbar^2}{2m}\frac{\nabla^2\psi}{\psi} + V(x) = i\hbar\frac{f'}{f} \tag{3.2}$$

$$f' = -\frac{iE}{\hbar}f \qquad -\frac{\hbar^2}{2m}\nabla^2\psi + V(x)\psi = E\psi \tag{3.3}$$

The wave function encodes the probability amplitude at a given time. Noting that $\hbar$ is called the 'reduced Planck constant', which is $\hbar = \frac{h}{2\pi}$, where h is the Planck constant $h \approx 6.626 \times 10^{-34}$ J·s, giving $\hbar \approx 1.055 \times 10^{-34}$ J·s.

For the 1D case this is defined as $\Psi(x,t)$ where $x \in r : \{x, y, z\}$, and describes the probability that the particle lies in the spatial window $[x_1, x_2]$:

$$P = \int_{x_1}^{x_2} \Psi^*(x,t)\,\Psi(x,t)\,dx = \int_{x_1}^{x_2} |\Psi(x,t)|^2\,dx \tag{3.4}$$

In order to extract logical information, $\Psi(x,t)$ is normalized to:

$$\int_{-\infty}^{\infty} |\Psi(x,t)|^2\,dx = 1 \tag{3.5}$$

This ensures the chance of finding a particle at (x_n, t_n) is always 100%, thus creating a physically-motivated probability distribution of 'where' the particle would be with respect to the operators applied (LibreTexts Mathematics, 2021).

3.1 Quantum Operators & Measurables

To measure the physical observables of a quantum mechanical system, one can apply a corresponding set of differential operators to the wave function, each of which yield an 'eigen-value problem' whose solutions define the possible valid states of the system.

Within the scope of this work we will not completely derive all the operators for the sake of brevity; though we provide a table of reference for the reader:

Observable	**Quantum Operator**	**Classical Operator**
Position	$\hat{x}$	x
Momentum	$\hat{P}_x = -i\hbar\,\partial_x$	$p^2/2m$
Kinetic Energy	$\hat{T}_x = -\dfrac{\hbar^2}{2m}\,\partial_x^2$	$\frac{1}{2}mv^2$
Energy	$\hat{H} = -\dfrac{\hbar^2}{2m}\,\partial_x^2 + V(x)$	$\frac{1}{2}mv^2 + V(x)$

Table 3.1: Taken and modified version of table found in (Weichmann, 2024)[1]

In classical mechanics, observables such as position and momentum are explicit functions of time that can be directly evaluated at any point in the system. In quantum mechanics however, the state of the system is encoded within the wave function $\Psi(x,t)$, and physical quantities can only be extracted by acting on that wave function with a corresponding differential operator.

This is done since the wave function itself is a complex-valued quantity and thus has no direct physical interpretation on its own; it is only through the action of an operator on Ψ that a physically meaningful quantity is produced. The operators are thus written as differential expressions in order to extract the rate of change of the wave function with respect to the quantity being measured, since it is precisely this rate of change that encodes the physical observable. This in turn means that applying an operator to the wave function transforms the problem into an eigenvalue equation, where the eigenvalues are the physically measurable quantities and the eigen-functions are the valid states of the system as mentioned previously.

This is important to note as we progress through this work, since rather than solving for an explicit trajectory as in the classical case, we will instead be constructing the appropriate Hamiltonian operator $\hat{H}$ for each system and solving $\hat{H}\Psi = E\Psi$, where the structure of $\hat{H}$ itself fully determines the dynamics and observable states of the system.

Chapter 4

Quantum Harmonic Oscillator

For quantum systems, the traditional harmonic oscillator is not viable as it does not account for many of the atomic properties found in quantum systems; and thus we modify the original system by the following:

1. Implement discrete energy level solutions, in accordance with experimental observations of atomic and molecular spectra, which show that bound systems absorb and emit energy in fixed quanta rather than continuously (Furman University, 2026).

2. Address the invalid notion of a true zero-energy rest state, since quantum particles cannot be simultaneously localized with zero momentum due to the Heisenberg uncertainty principle, preventing a perfectly stationary ground state (Wikipedia contributors, 2026c).

3. Construct non-deterministic trajectories consistent with the probabilistic interpretation of quantum mechanics, as formalized by the Born rule, where only probability amplitudes evolve deterministically rather than definite classical paths (Wikipedia contributors, 2026c).

To implement these changes, we first rewrite the classical harmonic oscillator Hamiltonian, substituting $k = m\omega^2$ for the spring constant:

$$H = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}m\omega^2 x^2 = \begin{cases} \dot{x} = v \\ \dot{v} = -\omega^2 x \end{cases} \tag{4.1}$$

Thus substituting the new Hamiltonian into $\Psi(x)$ gives the following eigen-value equation:

$$H\Psi = E\Psi \tag{4.2}$$

$$-\frac{\hbar^2}{2m}\frac{d^2}{dx^2}\psi + \frac{1}{2}m\omega^2 x^2\,\psi = E\psi \tag{4.3}$$

Rearranging the equation highlights a 2^{nd} order differential system, which we solve:

$$\frac{d^2\Psi}{dx^2} = \left[-\frac{2mE}{\hbar^2} + \frac{m^2\omega^2 x^2}{\hbar^2}\right]\Psi \tag{4.4}$$

Lemma: Note that in this analysis we will use ordinary differential equations due to the analysis being in only one spatial dimension. For any equations that utilize Partial Differential Equations, the reader may assume the context of the analysis is in dimensions greater than one; unless stated otherwise.

Since this system is considerably more complex and harder to solve either by ansatz or by solving for the eigen-values & vectors, we employ non-dimensionalization to simplify the co-factors & terms of the system.

We define $x = x_0 y$, where (y) is the dimensionless variable and (x_0) is the 'scaling parameter':

$$\frac{d}{dx}\left[\, x = x_0 y \,\right] \implies dx = x_0\, dy \tag{4.5}$$

$$\frac{d^2\Psi}{dx^2} = \frac{1}{x_0^2}\frac{d^2\Psi}{dy^2} \implies \frac{1}{x_0^2}\frac{d^2\Psi}{dy^2} = \left[-\frac{2mE}{\hbar^2} + \frac{m^2\omega^2(x_0 y)^2}{\hbar^2}\right]\Psi \tag{4.6}$$

$$d_y^2\,\Psi = \left[-\frac{2mEx_0^2}{\hbar^2} + \frac{m^2\omega^2 x_0^4}{\hbar^2}y^2\right]\Psi \tag{4.7}$$

Setting $\dfrac{m^2\omega^2 x_0^4}{\hbar^2} = 1$:

$$(m\omega x_0^2)^2 = \hbar^2 \implies x_0 = \sqrt{\frac{\hbar}{m\omega}} \tag{4.8}$$

Note: $\Psi(x) = u(y)$ (not required but clean as the entire system is non-dimensional).

Energy term:

$$\frac{2mE}{\hbar^2}(x_0^2) = \frac{2mE}{\hbar^2}\cdot\frac{\hbar}{m\omega} = \frac{2E}{\hbar\omega} \tag{4.9}$$

Therefore, assume $\varepsilon = \dfrac{2E}{\hbar\omega}\dots$

$$d_y^2\, u + (\varepsilon - y^2)u = 0 \tag{4.10}$$

where for $\lim_{y\to\pm\infty} u(y) = 0$, the (ε) term drops.

The resulting equation has solutions only at specific points that satisfy $u = u(y)$ that have discrete values of (ε).

To find a solution analytically we look at the behaviour of the system as u gains larger y values where $(y^2 \gg \varepsilon)$:

$$\frac{d^2u}{dy^2} - y^2 u = 0 \tag{4.11}$$

Ansatz: $u(y) = e^{ay^2} \implies u'(y) = 2aye^{ay^2}, \quad u''(y) = 4a^2y^2e^{ay^2}$

$$4a^2y^2e^{ay^2} - y^2e^{ay^2} = 0 \tag{4.12}$$

$$y^2e^{ay^2}\left[4a^2 - 1\right] = 0 \implies 4a^2 = 1 \implies a = \pm\frac{1}{2} \tag{4.13}$$

Where for large y, $u(y)$ changes at a rate of $e^{-y^2/2}$ due to the condition that the wave function must be normalized and at (∞) a growth is implausible and so the negative term dominates:
Ansatz:

$$u(y) = H(y)\, e^{-y^2/2} \tag{4.14}$$

Applying this to the original equation gives the following linear ODE without a 2nd order term:

$$u'(y) = H'(y)\, e^{-y^2/2} - H(y)\, y\, e^{-y^2/2} \tag{4.15}$$

$$u''(y) = H''(y)\, e^{-y^2/2} - 2y\, H'(y)\, e^{-y^2/2} + (y^2 - 1)H\, e^{-y^2/2} \tag{4.16}$$

Therefore:

$$H'' - 2yH' + H\varepsilon - H = 0 \tag{4.17}$$

To evaluate the $H(y)$ polynomial term, we assume a power-series as the solution for our ansatz due to the system having non-constant coefficients.
Ansatz:

$$H(y) = \sum_{j=0}^{\infty} a_j\, y^j \tag{4.18}$$

$$H'(y) = \sum_{j=0}^{\infty} j\, a_j\, y^{j-1} \tag{4.19}$$

$$H''(y) = \sum_{j=0}^{\infty} j(j-1)\, a_j\, y^{j-2} \tag{4.20}$$

Substituting this into our linear system $H'' - 2yH' + H\varepsilon - H = 0$:

$$\left[\sum_{j=0}^{\infty} j(j-1)\, a_j\, y^{j-2}\right] - \left[\sum_{j=0}^{\infty} 2j\, a_j\, y^j\right] + \left[\sum_{j=0}^{\infty} a_j\, y^j\right](\varepsilon - 1) = 0 \tag{4.21}$$

Since we need all the sum terms to have the same exponent of (y^j), we perform a substitution: $k = j - 2$, thus $j = k + 2$.
For the first term:

$$\sum_{j=0}^{\infty} j(j-1)\, a_j\, y^{j-2} = \sum_{k=0}^{\infty} (k+2)(k+1)\, a_{k+2}\, y^k \tag{4.22}$$

For the full equation:

$$\sum_{j=0}^{\infty} \left[(j+2)(j+1)\, a_{j+2} - 2j\, a_j + (\varepsilon - 1)\, a_j\right] y^j = 0 \tag{4.23}$$

This creates an infinite series. To solve this, all coefficients must equal zero:

$$(j+2)(j+1)\,a_{j+2} - 2j\,a_j + (\varepsilon - 1)\,a_j = 0 \tag{4.24}$$

$$(1) \qquad (j+2)(j+1)\,a_{j+2} - (2j + 1 - \varepsilon)\,a_j = 0 \tag{4.25}$$

$$(j+2)(j+1)\,a_{j+2} = (2j + 1 - \varepsilon)\,a_j \tag{4.26}$$

$$(2) \qquad a_{j+2} = \frac{(2j+1-\varepsilon)\,a_j}{(j+2)(j+1)} \tag{4.27}$$

Plugging back into (1):

$$(2j+1-\varepsilon)\,a_j - (2j+1-\varepsilon)\,a_j = 0 \tag{4.28}$$

Since this recursive relationship can go on indefinitely, the function $H(y)$ behaves like our $u(y) = e^{y^2}$ term which we stated could not occur due to normalization of the wave function.

To fix this we must find what is the terminating term of $(2j + 1 - \varepsilon)$ that has a concrete end.

Assuming the series runs for $j = n$ counts:

$$2n + 1 - \varepsilon = 0 \tag{4.29}$$

Substituting ($\varepsilon = 2E/\hbar\omega$):

$$2n + 1 - \frac{2E}{\hbar\omega} = 0 \tag{4.30}$$

$$\boxed{E_n = \left[n + \frac{1}{2}\right]\hbar\omega, \qquad n = 0, 1, 2, \ldots} \tag{4.31}$$

Evaluating the recurrence at $j = n = 0$ and $j = n = 1$:

$$a_{j+2} = \left.\frac{(2j+1-\varepsilon)\,a_j}{(j+2)(j+1)}\right|_{j=n=0} = \frac{(1-\varepsilon)\,a_0}{2} = 0 \tag{4.32}$$

$$a_{j+2} = \left.\frac{(2j+1-\varepsilon)\,a_j}{(j+2)(j+1)}\right|_{j=n=1,\,\varepsilon=3} = \frac{(3-3)\,a_1}{(3)(2)} = 0 \tag{4.33}$$

where $H_0(y) = 1$, $H_1(y) = 2y$, ...

This form of solutions is a known polynomial term called the Hermite polynomial of the form(Wikipedia contributors, 2025):

$$\mathfrak{H}_n(x) = (-1)^n e^{x^2} \frac{d^n}{dx^n} e^{-x^2} \tag{4.34}$$

Where a side-by-side comparison validates this:

$$\begin{array}{ll} \mathfrak{H}_0(x) = 1 & H_0(y) = 1 \\ \mathfrak{H}_1(x) = 2x & H_1(y) = 2y \\ \vdots & \vdots \end{array}$$

This shows that the series is consistent with the postulates of quantum mechanics when E_n constrains the $(2j+1-\varepsilon)$ term so that the series converges. This convergence occurs only for values of the form $\left(n+\frac{1}{2}\right)$, which correspond to the quantum number. As a result, the wavefunction is restricted to discrete solutions. Which is different from the classical harmonic oscillator, which allows a continuous range of energies over $\mathbb{R}^+$, whereas the quantum harmonic oscillator is quantized and only allows specific solutions described by Hermite polynomials.

Substituting this into our original dimensionless wave function:

$$u(y) = H(y)\, e^{-y^2/2} = N_n H_n(y)\, e^{-y^2/2} \tag{4.35}$$

To then finally write the physically motivated wave function in ordinary dimensions, we now reverse the non-dimensionalization:

$$y = \sqrt{\frac{m\omega}{\hbar}}\, x_0 \qquad \Psi(x) = u(y) \tag{4.36}$$

$$\Psi_n(x) = N_n H_n\left(\sqrt{\frac{m\omega}{\hbar}}\, x\right) e^{-\frac{m\omega}{2\hbar}x^2} \tag{4.37}$$

Then normalizing this $\int_{-\infty}^{\infty} |\Psi_n(x)|^2\, dx = 1 \ldots$

$$N_n = \left[\frac{m\omega}{\pi\hbar}\right]^{1/4} \frac{1}{\sqrt{2^n n!}} \qquad \sim \text{known property for Hermite polynomials} \ldots \tag{4.38}$$

$$\Psi_n(x) = \left[\frac{m\omega}{\pi\hbar}\right]^{1/4} \frac{1}{\sqrt{2^n n!}}\, H_n\left[\sqrt{\frac{m\omega}{\hbar}}\, x\right] e^{-\frac{m\omega}{2\hbar}x^2} \tag{4.39}$$

4.1 Time-Dependent Solution

To then solve for a time-dependent solution which then can be plotted with respect to time, we solve the time-dependent Schrödinger equation (TDSE):

$$i\hbar \frac{\partial \Psi(x,t)}{\partial t} = \hat{H}\Psi(x,t) \tag{4.40}$$

where $\hat{H} = -\dfrac{\hbar^2}{2m}\dfrac{d^2}{dx^2}\psi + \dfrac{1}{2}m\omega^2 x^2\psi$ is not time dependent; thus we can split the wave function into two parts (spatial & temporal):

$$\Psi(x,t) = \Psi(x)\,\phi(t) \tag{4.41}$$

Substituting into the TDSE:

$$i\hbar \frac{\partial \Psi(x,t)}{\partial t} = \hat{H}\Psi(x,t) \tag{4.42}$$

$$i\hbar\,\Psi(x)\,\frac{d\phi(t)}{dt} = \phi(t)\,\hat{H}\Psi(x) \tag{4.43}$$

$$i\hbar\,\frac{1}{\phi(t)}\,\frac{d\phi(t)}{dt} = \frac{1}{\Psi(x)}\,\hat{H}\Psi(x) \tag{4.44}$$

Here since the RHS is solely dependent on space (x) and the LHS is solely dependent on time (t) they must be equal:

$$i\hbar\,\frac{d\phi(t)}{dt} = \hat{H}, \qquad \hat{H}\Psi = E\Psi \tag{4.45}$$

$$i\hbar\,\frac{d\phi}{dt} = E_n\,\phi \tag{4.46}$$

Which is a simple separable system:

$$i\hbar \int \frac{1}{\phi}\,d\phi = \int E_n\,dt \tag{4.47}$$

$$i\hbar\,\ln|\phi| = E_n t + C \tag{4.48}$$

$$\ln|\phi| = \frac{E_n t}{i\hbar} + \frac{C}{i\hbar} \tag{4.49}$$

$$\phi(t) = e^{-iE_n t/\hbar + C/i\hbar} \tag{4.50}$$

where as $t \to \pm\infty$, $C/i\hbar$ gets dominated and is absorbed:

$$\phi(t) = e^{-iE_n t/\hbar} \tag{4.51}$$

Which means $\Psi(x,t) = \psi_n(x)\,e^{-iE_n t/\hbar}$ is stationary for $\forall\,t$ if $|\phi(t)|^2 = 1$, thus only truly depending on the spatial state.

This is important to note as systems such as Hydrogen electron orbitals are traditionally time-independent / spatially dominated solutions and thus don't require solving for a generic time-evolving solution unlike the CHO.

This is seen by the fact that the potential energy for an electron in an atom is given by:

$$V(x) = -\frac{Ze^2}{4\pi\varepsilon_0 r} \tag{4.52}$$

where:

- r : radius to nucleus
- Ze^2 : charge of nucleus
- ε_0 : constant of electric permittivity

Which is not time-dependent, thus creating a similar system that will have a spatially dominated wave function.

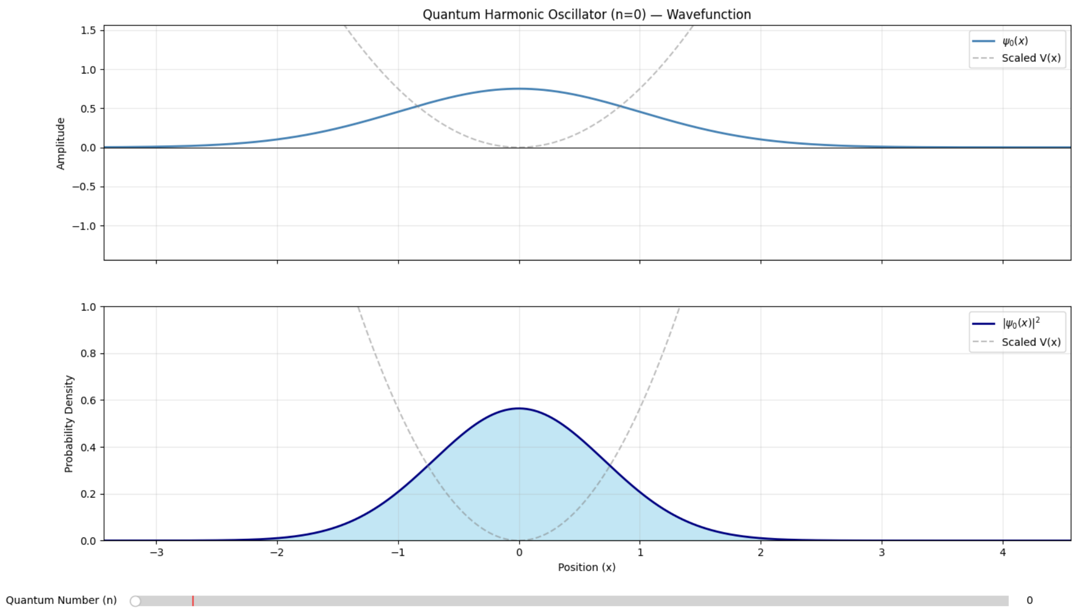

Figure 4.1: A quantum harmonic oscillator set at n=0 quantum principle number.

4.2 Implications & Asymptotic Analysis

These results highlights the differences from the classical harmonic oscillator we saw in the prior chapter. Where unlike how the classical system allowed the energy to take any value in $\mathbb{R}^+$ and the trajectory of the particle was fully determined by its initial conditions, the quantum system can only permit discrete energy values $E_n = [n+\frac{1}{2}]\hbar\omega$, each corresponding to a unique eigenfunction $\Psi_n(x)$ of the system. This is a direct consequence of the non-constant coefficient structure of the quantum equation, which unlike the classical system which could not be solved by the eigenvalue methods of a linear matrix system. Furthermore it is noted that even at $n = 0$ the system carries a zero-point energy $E_0 = \frac{1}{2}\hbar\omega$, meaning the quantum oscillator can never truly be at rest regardless of its state, which has no analogue in the classical picture. Similarly the probability distribution $|\Psi_n(x)|^2$ at low n is concentrated near the equilibrium position rather than the turning points, directly contradicting the classical distribution established in the prior chapter. As the quantum number (n) is increased however, the wavefunction $\Psi_n(x)$ develops an increasing number of oscillatory nodes across the classically accessible region, with the nodal spacing becoming progressively finer and the large-scale average of $|\Psi_n(x)|^2$ beginning to flatten across the domain. In the limit of large (n) this average converges onto a distribution that closely follows the form of the potential $V(x) = \frac{1}{2}m\omega^2x^2$, concentrating near the turning points $x = \pm A$ in the same manner as the classical oscillator, since the energy spacing $\hbar\omega$ becomes negligible relative to $E_n \approx n\hbar\omega$ and the rapid oscillations wash out into an effective continuum. Thus the quantum oscillator, whilst fundamentally discrete, asymptotically behaves similarly to its classical counterpart.

Chapter 5

The Hydrogen Atom

For many harmonic systems, we have seen that their fundamental properties can change how they propagate. However, we have primarily focused on one-body systems, where energy conservation involves only the system itself and its environment.

We now apply the methods & properties previously discussed to build an equation for the probabilistic-dynamics of a two body system; notably the hydrogen atom. To do this we first must modify the traditional 1D-Schrödinger equation to account for the two-bodied nature of the system.

5.1 Schrödinger Equation & Relative Coordinates

Since the hydrogen atom has one proton and one electron, the Hamiltonian needs to account for both the electron's and proton's kinetic and potential energy, thus:

$$\hat{H} = -\frac{\hbar^2}{2m_e}\nabla_e^2 - \frac{\hbar^2}{2m_p}\nabla_p^2 + V(|r_e - r_p|) \tag{5.1}$$

This specific expression is often too difficult to solve directly due to the potentials of the system being coupled to each other; making it difficult to cleanly separate them.

Instead we perform a change of variables that instead of tracking each particle individually, instead looks at the 'centered' weight of the system. This helps by reducing the two-body problem to a relative one-body system.

We define the following change of coordinates:

1. R: the weighted center of the whole system
2. r: the distance & direction of the electron relative to the proton

This is done by assuming the proton body always stays at the coordinate-origin of the electron-proton system at all times. This helps us isolate electrostatic & kinetic forces from the electron to the proton; treating the proton as the reference point.

$$R = \frac{m_e r_e + m_p r_p}{M}, \qquad r = r_e - r_p, \qquad M = m_e + m_p. \tag{5.2}$$

This simplifies the problem as:

$$\hat{H} = -\frac{\hbar^2}{2m_e}\nabla_e^2 - \frac{\hbar^2}{2m_p}\nabla_p^2 + V(|r_e - r_p|) \tag{5.3}$$

$$\nabla_e = \frac{\partial R}{\partial r_e}\nabla_R + \frac{\partial r}{\partial r_e}\nabla_r \qquad \nabla_p = \frac{\partial R}{\partial r_p}\nabla_R + \frac{\partial r}{\partial r_p}\nabla_r \tag{5.4}$$

$$\begin{aligned}\nabla_e^2 &= \left(\frac{m_e}{M}\nabla_R + \nabla_r\right)\cdot\left(\frac{m_e}{M}\nabla_R + \nabla_r\right)\\ &= \left[\frac{m_e}{M}\right]^2\nabla_R^2 + \frac{2m_e}{M}(\nabla_R\cdot\nabla_r) + \nabla_r^2\end{aligned} \tag{5.5}$$

$$\begin{aligned}\nabla_p^2 &= \left(\frac{m_p}{M}\nabla_R - \nabla_r\right)\cdot\left(\frac{m_p}{M}\nabla_R - \nabla_r\right)\\ &= \left[\frac{m_p}{M}\right]^2\nabla_R^2 - \frac{2m_p}{M}(\nabla_R\cdot\nabla_r) + \nabla_r^2\end{aligned} \tag{5.6}$$

With $\hat{H} = T + V$, the kinetic term expands (noting that the cross terms $(\nabla_R \cdot \nabla_r)$ cancel):

$$\begin{aligned}T = &-\frac{\hbar^2}{2m_e}\left[\frac{m_e^2}{M^2}\nabla_R^2 + \frac{2m_e}{M}(\nabla_e\cdot\nabla_r) + \nabla_r^2\right]\\ &-\frac{\hbar^2}{2m_p}\left[\frac{m_p^2}{M^2}\nabla_R^2 - \frac{2m_p}{M}(\nabla_R\cdot\nabla_r) + \nabla_r^2\right]\\ = &-\frac{\hbar^2}{2}\left[\left(\frac{m_e}{M^2} + \frac{m_p}{M^2}\right)\nabla_R^2 + \left(\frac{1}{m_e} + \frac{1}{m_p}\right)\nabla_r^2\right]\end{aligned} \tag{5.7}$$

where:

$$\frac{m_e + m_p}{M^2} = \frac{1}{M} \qquad \frac{1}{m_e} + \frac{1}{m_p} = \frac{M}{m_e m_p} = \frac{1}{\mu} \tag{5.8}$$

Therefore:

$$\hat{H} = -\frac{\hbar^2}{2M}\nabla_R^2 - \frac{\hbar^2}{2\mu}\nabla_r^2 + V(r) \tag{5.9}$$

where (μ) is the reduced mass of the proton–electron system, defined by $\mu^{-1} = m_e^{-1} + m_p^{-1}$. Thus applying that to the wave equation:

$$\left[-\frac{\hbar^2}{2M}\nabla_R^2 - \frac{\hbar^2}{2\mu}\nabla_r^2 + V(r)\right]\Psi(R,r) = E\Psi(R,r) \tag{5.10}$$

Because $V(r)$ is de-coupled from (R), the system can be written as two separate wave functions ψ and φ:

$$\Psi(R,r) = \psi(R)\,\varphi(r) \tag{5.11}$$

$$\frac{1}{\psi(R)}\left[-\frac{\hbar^2}{2M}\nabla_R^2\psi(R)\right] + \frac{1}{\varphi(r)}\left[-\frac{\hbar^2}{2\mu}\nabla_r^2\varphi(r) + V(r)\varphi(r)\right] = E \tag{5.12}$$

This allows us to completely ignore the (R) focused part of the system as the Coulomb potential:

$$V(r) = -\frac{Ze^2}{4\pi\varepsilon_0 r} \tag{5.13}$$

similarly only depends on (r), thus allowing us to reduce our problem to the following:

$$-\frac{\hbar^2}{2\mu}\nabla_r^2\,\psi(r) + V(r)\psi(r) = E\psi(r) \tag{5.14}$$

5.2 Solving For The Radial Equation

A key factor about solving for the atom's wave-function is to utilize the fact that the entire system is within a spherical region. This allows us to simplify the tracking of the system even more by performing a change of variables to spherical coordinates:

$$x = r\sin\theta\cos\phi \qquad y = r\sin\theta\sin\phi \qquad z = r\cos\phi \tag{5.15}$$

$$J = r^2\sin\theta\,dr\,d\theta\,d\phi \tag{5.16}$$

Where we perform:

$$\nabla^2 = \frac{1}{r^2}\frac{\partial}{\partial r}\left[r^2\frac{\partial}{\partial r}\right] + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left[\sin\theta\frac{\partial}{\partial\theta}\right] + \frac{1}{r^2\sin^2\theta}\frac{\partial^2}{\partial\phi^2} \tag{5.17}$$

Similar to the previous section, we analyse the variable-dependency of the wave-function to split the total function into independent parts. The Laplacian separates into two primary types: angular dependency (rotational & azimuthal) and radial dependency. Thus we write:

$$\psi(r,\theta,\phi) = R(r)\,Y(\theta,\phi) \tag{5.18}$$

Substituting $\psi = R(r)\,Y(\theta,\phi)$ into the Schrödinger equation in spherical coordinates and dividing through by $R(r)Y(\theta,\phi)$ separates the equation into distinct radial and angular parts, each of which must equal the same separation constant. The angular equation (2) reduces to a form of the associated Legendre equation. Requiring the solutions to be finite and single-valued on the sphere restricts the separation constant to the form $\ell(\ell+1)$, where $(\ell = 0, 1, 2, \ldots)$. This is a standard result from the angular analysis, which we do not reproduce here since our focus is on the radial component (1).

$$(1)\quad \frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right) + \left\{\frac{2\mu}{\hbar^2}(E - V(r)) - \frac{\ell(\ell+1)}{r^2}\right\}R = 0 \tag{5.19}$$

$$(2)\quad \frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial Y}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2 Y}{\partial\phi^2} + \ell(\ell+1)Y = 0 \tag{5.20}$$

The angular component (2) is a known equation whose solutions are the spherical harmonics $Y_\ell^m(\theta,\phi)$, parameterized by the orbital quantum number (ℓ) and magnetic quantum number (m). Since we are constructing radial probability distributions, the angular component integrates out cleanly due to the normalization of the spherical harmonics:

$$\int_0^{2\pi}\int_0^{\pi} |Y_\ell^m(\theta,\phi)|^2 \sin\theta\, d\theta\, d\phi = 1 \tag{5.21}$$

This allows us to reduce our analysis entirely to the radial component (1). We now solve this directly.

Implementing $V(r) = -Ze^2/4\pi\varepsilon_0 r$ and $\lambda_1 = \ell(\ell+1)$:

$$\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right) + \left\{\frac{2\mu}{\hbar^2}\left[E + \frac{Ze^2}{4\pi\varepsilon_0 r}\right] - \frac{\ell(\ell+1)}{r^2}\right\} R = 0 \tag{5.22}$$

Now we non-dimensionalize this system by the following change in variables:

$$\rho = \frac{\sqrt{8\mu|E|}}{\hbar}\, r \qquad u(\rho) = r\, R(r) \tag{5.23}$$

which after simplification gives:

$$\frac{d^2u}{d\rho^2} + \left\{\frac{\alpha}{\rho} - \frac{1}{4} - \frac{\ell(\ell+1)}{\rho^2}\right\} u = 0 \tag{5.24}$$

where:

$$\alpha = \frac{Ze^2}{4\pi\varepsilon_0\hbar}\sqrt{\frac{\mu}{2|E|}} \tag{5.25}$$

Now rewriting the system as $u = u(\rho)$ for large (ρ):

$$\frac{d^2u}{d\rho^2} - \frac{1}{4}u = 0 \tag{5.26}$$

Writing this as a coupled linear system...

$$\begin{cases} d_\rho u = v \\ d_\rho v = \dfrac{1}{4}u \end{cases} \implies \begin{bmatrix} u' \\ v' \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ \frac{1}{4} & 0 \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} \tag{5.27}$$

$$\lambda^2 - \frac{1}{4} = 0 \implies \lambda = \pm\frac{1}{2} \tag{5.28}$$

Thus has two linearly independent solutions:

$$u(\rho) = \begin{cases} e^{\frac{1}{2}\rho} \\ e^{-\frac{1}{2}\rho} \end{cases} \tag{5.29}$$

Since the wave function must remain normalizable, the growing term $e^{\rho/2}$ is not physically valid and so we take $u(\rho) \sim e^{-\rho/2}$ as the asymptotic behaviour. Similarly near the origin, assuming a power-series ansatz $u \approx \rho^s$ and dropping sub-dominant terms:

$$s(s-1) - \ell(\ell+1) = 0 \implies s = \begin{cases} \ell + 1 \\ -\ell \end{cases} \tag{5.30}$$

Since $\lim_{\rho\to 0} u(\rho) = 0$ is required for $R(r)$ to remain finite at the origin, we take $s = \ell+1$. Combining both asymptotic behaviours we write:

$$u(\rho) = \rho^{\ell+1} e^{-\rho/2} F(\rho) \tag{5.31}$$

Substituting this into the dimensionless radial equation gives the following linear system for $F(\rho)$:

$$\frac{d^2F}{d\rho^2} + \left(\frac{2(\ell+1)}{\rho} - 1\right)\frac{dF}{d\rho} + \frac{\alpha - (\ell+1)}{\rho} F = 0 \tag{5.32}$$

Once again we apply a power-series ansatz:

$$F(\rho) = \sum_{k=0}^{\infty} a_k \, \rho^k \tag{5.33}$$

to get the following recursive relationship:

$$a_{k+1} = \frac{1 + \ell + k - \alpha}{(k+1)(k + 2(\ell+1))} \, a_k \tag{5.34}$$

Since for large (k) this series grows as e^{ρ}, which would violate normalization, the series must terminate. This is done by requiring the numerator to vanish at some $k = n'$:

$$\alpha = 1 + \ell + n', \qquad n' = 0, 1, 2, \ldots \tag{5.35}$$

Defining the principal quantum number $n = 1 + \ell + n'$ and solving for E_n similarly to the quantum harmonic oscillator:

$$\frac{Ze^2}{4\pi\varepsilon_0\hbar}\sqrt{\frac{\mu}{2|E|}} = n \implies \sqrt{\frac{\mu}{2|E|}} = \frac{4\pi\varepsilon_0\hbar n}{Ze^2} \implies |E| = \frac{\mu Z^2 e^4}{2(4\pi\varepsilon_0)^2\hbar^2 n^2} \tag{5.36}$$

$$\boxed{E_n = -\frac{\mu Z^2 e^4}{2(4\pi\varepsilon_0)^2\hbar^2 n^2}, \qquad \Psi_{n\ell} = \begin{cases} n = 1, 2, 3, \ldots \\ \ell = 0, 1, \ldots, n-1 \end{cases}} \tag{5.37}$$

The terminated $F(\rho)$ polynomial is a known function called the associated Laguerre polynomial (Wikipedia contributors, 2026b), and together with the asymptotic factors gives the full radial solution:

$$u(\rho) = \rho^{\ell+1} e^{-\rho/2} F(\rho), \qquad F(\rho) \to L^{2\ell+1}_{n-\ell-1}(\rho) \implies R_{n\ell}(r) = \frac{u(\rho)}{r} \tag{5.38}$$

$$R_{n\ell}(r) = \frac{u(\rho)}{r} = N_{n\ell}\, \rho^{\ell}\, e^{-\rho/2}\, L^{2\ell+1}_{n-\ell-1}(\rho) \tag{5.39}$$

where $N_{n\ell}$ is the normalization constant found by:

$$\int_0^{\infty} |R_{n\ell}(r)|^2 \, r^2 \, dr = 1 \tag{5.40}$$

5.3 Radial Probability Distribution

From the radial solution $R_{n\ell}(r)$ derived in the previous section, we can now model the probability of finding the electron at a given radius (r) from the nucleus. This is defined as the radial probability distribution $P(r)$, obtained by taking the square modulus of the radial solution weighted by the spherical volume element $r^2 \sin\theta \, dr \, d\theta \, d\phi$, and integrating over the angular components:

$$P(r) = r^2 \, |R_{n\ell}(r)|^2 \tag{5.41}$$

where the r^2 factor comes directly from the volume element after integration over (θ) and (ϕ). This is an important distinction from $|R_{n\ell}|^2$ itself, since whilst the radial function is maximal at the origin for $\ell = 0$ states, the r^2 factor suppresses the distribution near $r = 0$, thus the most probable radius is always found at a finite distance from the nucleus. For the ground state $n = 1$, $\ell = 0$ this is exactly the Bohr radius a_0, seen directly from the peak of $P(r)$.

Angular Momentum & The Classical Limit

The role of ℓ is most clearly seen by varying it while keeping n fixed, so that the energy E_n remains unchanged across the comparison. For a given principal quantum number n, the orbital quantum number takes values $\ell = 0, 1, \ldots, n-1$, which has a coupled effect on the radial probability distribution $P(r) = r^2|R_{n\ell}(r)|^2$. At $\ell = 0$, the s-state exhibits $n-1$ radial nodes and a distribution that is broadly spread across the classically allowed region, typically with several distinct peaks. As ℓ increases toward its maximum value $\ell = n-1$, the number of radial nodes decreases as $n - \ell - 1$, and the distribution becomes increasingly concentrated into a single dominant peak near $r \approx n^2 a_0$.

This behaviour has a classical counterpart present in newtonian mechanics. In a Coulomb potential, classical orbits at fixed energy E_n range from ellipses at low angular momentum to circular orbits at the maximum angular momentum. Circular motion confines the particle to a single radius, whereas the elliptical orbits explore a wide range of radial distances over a period. Thus the quantum state with $\ell = n-1$ is the analogue of the circular orbit: it carries the maximum angular momentum allowed for energy E_n, has no radial nodes, and produces a sharply peaked $P(r)$ near the classical orbital radius $n^2 a_0$. By contrast, the $\ell = 0$ state corresponds to a nearly radial trajectory, with the particle distributed across the full extent of the potential well.

Thus, in the limit $n \to \infty$, it is specifically the $\ell = n-1$ states that converge most directly to the classical picture of circular motion in a Coulomb potential. States with fixed ℓ, on the other hand, approach the classical limit more slowly and correspond instead to increasingly eccentric classical trajectories rather than circular ones.

With the following system we can then solve for any configuration of the orbitals. For instance the ground state $n = 1$, $\ell = 0$ (LibreTexts Mathematics, 2021):

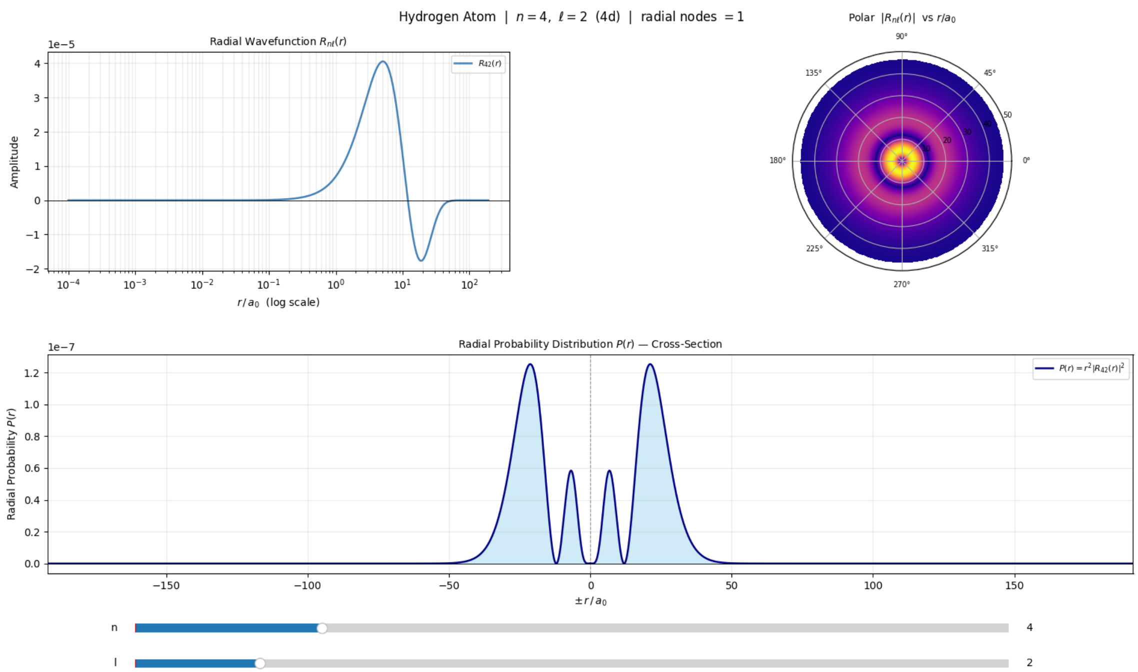

Figure 5.1: An electron orbital of a hydrogen atom with quantum numbers $n = 4$ and $\ell = 2$. The radial probability distribution is extended symmetrically into $r < 0$ to provide a side-view visualization of the orbital: the left side corresponds to the rear of the atom, while the right side represents the front.

$$R_{10}(r) = 2\left(\frac{1}{a_0}\right)^{3/2} e^{-r/a_0} \implies P(r) = \frac{4r^2}{a_0^3}\, e^{-2r/a_0} \tag{5.42}$$

which gives a single peak at the Bohr radius a_0 with no radial nodes. For the first excited state $n = 2$, $\ell = 1$:

$$R_{21}(r) = \frac{1}{\sqrt{24}}\left(\frac{1}{a_0}\right)^{3/2} \frac{r}{a_0}\, e^{-r/2a_0} \implies P(r) = \frac{r^4}{24a_0^5}\, e^{-r/a_0} \tag{5.43}$$

which introduces a broader distribution pushed further from the nucleus. For the higher excited state $n = 4$, $\ell = 2$:

$$R_{42}(r) = \frac{1}{96\sqrt{5}}\left(\frac{1}{a_0}\right)^{3/2}\left(\frac{r}{a_0}\right)^2\left(8 - \frac{r}{a_0}\right) e^{-r/4a_0} \implies P(r) = \frac{r^6}{46080\, a_0^7}\left(8 - \frac{r}{a_0}\right)^2 e^{-r/2a_0} \tag{5.44}$$

which introduces two radial nodes and a broader, more diffused distribution pushed further from the nucleus, consistent with the $n = 4$, $\ell = 2$ ($4d$) orbital. Based on these properties there are some important implications we can observe directly from our system that are of note.

5.4 Implications & Asymptotic Analysis

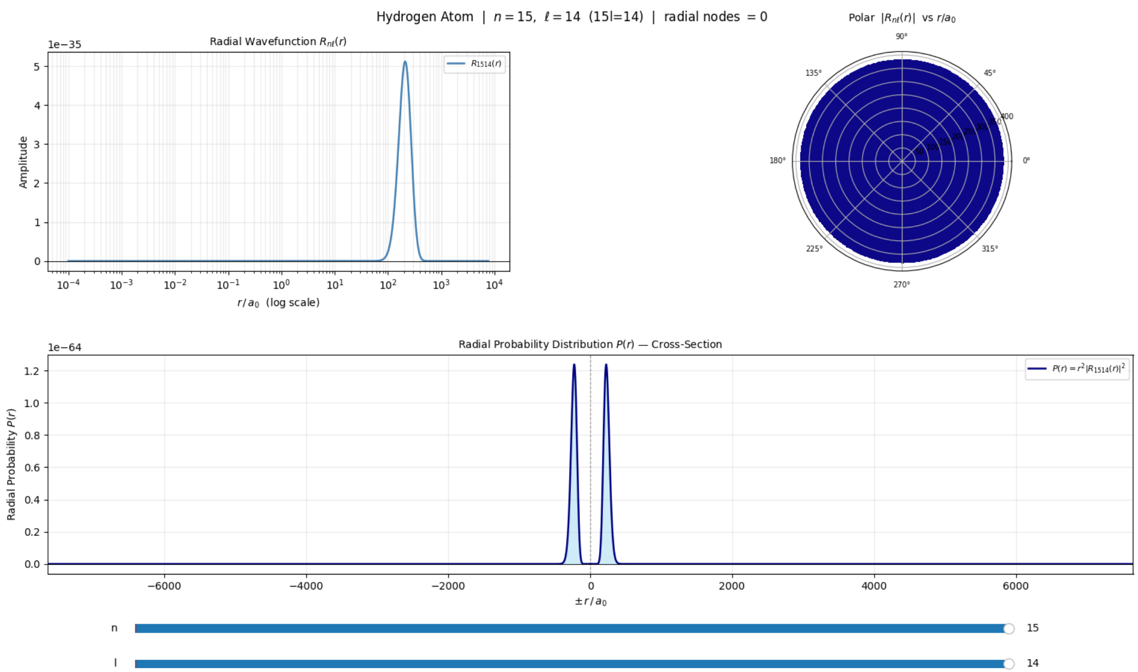

Figure 5.2: A hydrogen atom electron orbital set to $n = 15, \ell = 14$. The radial probability distribution is extended symmetrically into $r < 0$ to provide a side-view visualization of the orbital: the left side corresponds to the rear of the atom, while the right side represents the front. This highlights how the electron is only seen in one potential orbit around the nucleus, unlike the non-maximal states.

Lastly, as $n \to \infty$, the most probable radius scales as $n^2 a_0$, which follows directly from the peak of the radial probability distribution $P(r) = r^2 |R_{n\ell}(r)|^2$, whilst the distribution itself becomes increasingly spread out over a larger spatial region. At the same time, the $n - \ell - 1$ radial nodes of $R_{n\ell}(r)$ become more densely packed as the classically accessible region expands. The binding energy $E_n = -\frac{\mu Z^2 e^4}{2(4\pi\varepsilon_0)^2 \hbar^2 n^2}$ decreases as $1/n^2$, approaching zero from below, and in the large-n limit the system therefore approaches the ionization threshold $E = 0$, beyond which the electron is no longer bound (Coy, 1999).

In this threshold region, the de Broglie wavelength $\lambda = h/p$ of the electron becomes small relative to the spatial scale over which the Coulomb potential $V(r) = -Ze^2/4\pi\varepsilon_0 r$ varies, which is precisely the condition under which the probabilistic wave description reduces to a classical trajectory. The orbital dynamics of these highly excited states, commonly referred to as Rydberg states (Glab, 1997), are thus amenable to Newtonian analysis as a justified asymptotic limit that emerges naturally from the structure of the quantum solutions themselves.

Algorithms & Implementations

5.5 Classical Harmonic Oscillator: Implementation

To solve this numerically we follow the given algorithmic sequence:

1. Create linearly spaced time intervals between $(t_0 \to t_n)$
2. Solve the given initial value problem:

$$\begin{bmatrix} x' \\ v' \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -k/m & 0 \end{bmatrix} Y \tag{5.45}$$

Lemma: To perform this, we utilize the Dormand–Prince RK4(5) algorithm to create a temporally discretized approximation. This is an embedded adaptive Runge–Kutta method (Mathews & Fink, 2004) in which a fourth-order and a fifth-order update are computed simultaneously from the same set of six stage evaluations; their pointwise difference is then used as a local error estimate to automatically shrink or grow the step-size Δt so that the error is kept near a pre-defined fixed tolerance (Wikipedia contributors, 2026a).

Algorithm 1 Dormand–Prince RK4(5) Step

---

1. Given $y' = f(t, y)$, step size h, and (t_n, y_n).
2. Compute six stage slopes using the Dormand–Prince tableau:

$$k_i = f\left(t_n + c_i h,\ y_n + h \sum_{j<i} a_{ij} k_j\right), \quad i = 1, \ldots, 6.$$

3. Form the fourth- and fifth-order updates from the same stages:

$$y_{n+1}^{(r)} = y_n + h \sum_{i=1}^{6} b_i^{(r)} k_i, \quad r \in \{4, 5\}.$$

4. Error estimate:

$$e = \left\| y_{n+1}^{(5)} - y_{n+1}^{(4)} \right\|.$$

5. Step size control:

$$h_{\text{new}} = h \left(\frac{\text{tol}}{e}\right)^{1/5}.$$

6. Accept step if $e < \text{tol}$, otherwise reject and recompute with h_{new}.

5.6 Quantum Harmonic Oscillator: Implementation

To evaluate this numerically we follow the given algorithmic sequence:

1. Create a linearly spaced spatial grid over the domain ($x_0 \to x_n$)
2. Evaluate the normalized wavefunction directly from the known analytical solution:

$$\Psi_n(x) = N_n \, H_n(\alpha x) \, e^{-\alpha^2 x^2/2}, \qquad \alpha = \sqrt{\frac{m\omega}{\hbar}} \tag{5.46}$$

3. Compute the probability density $|\Psi_n(x)|^2$ from the evaluated wavefunction

Lemma: Unlike the classical harmonic oscillator which required numerical integration of an initial value problem, the quantum harmonic oscillator is evaluated directly from its closed form analytical solution. This is done by computing the Hermite polynomial H_n at each spatial point via the known recurrence relation, applying the Gaussian envelope $e^{-\alpha^2 x^2/2}$, and scaling by the normalization constant N_n. The slider over (n) re-evaluates the wavefunction and probability density at each new quantum number without requiring any additional numerical integration.

5.7 Hydrogen Atom: Implementation

To evaluate this numerically we follow the given algorithmic sequence:

1. Construct a linearly spaced radial grid over the domain ($r_{\min} \to r_{\max}$), where $r_{\max} = 2n^2 a_0(n+2)$ is chosen to capture the full orbital extent for any given (n)
2. Evaluate the normalized radial wavefunction directly from the known analytical solution:

$$R_{n\ell}(r) = N_{n\ell} \, e^{-\rho/2} \, \rho^\ell \, L_{n-\ell-1}^{2\ell+1}(\rho), \qquad \rho = \frac{2Zr}{na_0} \tag{5.47}$$

3. Compute the radial probability distribution $P(r) = r^2 |R_{n\ell}(r)|^2$ from the evaluated wavefunction

Lemma: Similarly to the quantum harmonic oscillator, the radial wavefunction is evaluated directly from its closed form analytical solution rather than through numerical integration. The associated Laguerre polynomial $L_{n-\ell-1}^{2\ell+1}$ is evaluated at each radial grid point via Horner's method applied to its known polynomial coefficients, after which the exponential envelope $e^{-\rho/2}$ and the centrifugal factor ρ^ℓ are applied and scaled by the normalization constant $N_{n\ell}$. The two sliders over (n) and (ℓ) re-evaluate the wavefunction and probability distribution at each new quantum number combination, with the radial domain automatically rescaling to $2n^2 a_0(n+2)$ to ensure the full orbital extent is always captured.

References

Coy, K. (1999). Hydrogen spectra ionization [Accessed: 2026-04-17]. https://web.phys.ksu.edu/vqmorig/tutorials/online/hydrogen/ionization.html

Furman University. (2026, February 17). *Understanding atomic spectra* [LibreTexts Chemistry]. https://chem.libretexts.org/@go/page/85153

Glab, W. L. (1997, July). What's a rydberg state, anyway? [physweb] [Accessed: 2026-04-17]. http://www.phys.ttu.edu/~gglab/rydberg_state.html

LibreTexts Engineering. (2022, May). Hilbert spaces. https://eng.libretexts.org/@go/page/22939

LibreTexts Mathematics. (2021, November). The Schrödinger equation. Retrieved April 15, 2026, from https://math.libretexts.org/@go/page/90439

LibreTexts Physics. (2025, March). The quantum harmonic oscillator. https://phys.libretexts.org/@go/page/4531

Mathews, J. H., & Fink, K. K. (2004). *Numerical methods using MATLAB* (4th ed.). Prentice-Hall.

Weichmann, M. (2024). CHM 305 – Lecture 6: The Postulates of Quantum Mechanics (Part 1) [Lecture notes]. https://weichman.princeton.edu/wp-content/uploads/2024/10/chm305_2024_lecture6.pdf

Wikipedia contributors. (2025). Hermite polynomials. https://en.wikipedia.org/wiki/Hermite_Polynomials

Wikipedia contributors. (2026a, February). *Dormand–prince method*. Wikipedia, The Free Encyclopedia. Retrieved April 17, 2026, from https://en.wikipedia.org/w/index.php?title=Dormand%E2%80%93Prince_method&oldid=1338750478

Wikipedia contributors. (2026b, February). Laguerre polynomials. Retrieved April 15, 2026, from https://en.wikipedia.org/w/index.php?title=Laguerre_polynomials&oldid=1339484117

Wikipedia contributors. (2026c, April 10). *Zero-point energy* [In Wikipedia, The Free Encyclopedia. Retrieved April 16, 2026]. https://en.wikipedia.org/w/index.php?title=Zero-point_energy&oldid=1348001184